

SUBODH KUMAR

PROFESSIONAL SUMMARY

Product-focused Backend Engineer with 4.5+ years of experience building scalable fintech, SaaS, and AI-powered systems. Strong expertise in backend architecture, payment integrations, API engineering, and production-grade application development, combined with hands-on experience in machine learning integration, RAG pipelines, FastAPI services, and AI-enabled workflows.

Experienced in designing high-volume transactional systems, predictive analytics integrations, conversational AI platforms, and deployment-ready backend services. Skilled at bridging software engineering and applied AI to create scalable, secure, and intelligent products.

CORE EXPERTISE

- Backend Systems & APIs
- AI-Powered Application Development
- Product Engineering & Ownership
- FastAPI & Laravel Development
- RAG Pipelines & Context-Aware Retrieval
- Payment Gateway & Fintech Integrations
- Predictive Analytics & ML Integration
- Secure & High-Availability Systems
- Scalable Architecture & Optimization
- REST APIs & Model Serving

Product Engineer | Software Engineer

Backend Systems | Applied AI | FastAPI | Laravel | Machine Learning

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PROFESSIONAL EXPERIENCE

Vayam Technologies Ltd.

August 2023 – Present

Product Engineer / Software Engineer

MaddoxPay Fintech Platform

Backend Systems & Product Ownership

- Designed and owned high-volume recharge and bill-payment systems handling 100K+ monthly transactions.
- Implemented automated payout systems processing ₹1M+ daily through Axis Bank integrations including NEFT, RTGS, and IMPS.
- Architected cross-vendor PAN booking systems processing 40K+ monthly secure transactions with reconciliation workflows.
- Optimized backend architecture, database schemas, and query execution, achieving nearly 2× faster response performance.
- Built admin analytics dashboards and operational monitoring systems for transaction tracking and business insights.

Applied AI / ML Integration

- Integrated predictive machine learning models into internal workflows for anomaly detection, analytics, and operational intelligence.
- Designed APIs for model serving and real-time prediction workflows.
- Worked on deployment-ready AI integrations with reliability-focused fallback handling.
- Supported AI-enabled backend automation and data-driven decision systems.

Security & Infrastructure

- Implemented MFA, geofencing, secure sessions, suspicious-login detection, and fraud prevention workflows.
- Integrated UIDAI-compliant biometric eKYC systems using Morpho and Mantra devices.
- Reduced platform downtime through backend reliability improvements and proactive infrastructure management.
- Managed production deployments, uptime monitoring, and backend services on Windows and CentOS environments.

Key Services

AEPS, DMT, MATM, Payout Systems, PAN Services, Payment Services, Recharge & Billing Systems

Flyde Digital Solutions

July 2022 – August 2023

Fullstack Engineer

- Delivered backend-focused full-stack applications with scalable API integrations and optimized workflows.

TECHNICAL SKILLS

Languages & Backend

- Python
- PHP (Laravel)
- SQL
- FastAPI
- REST APIs
- JWT Authentication

AI / ML & Applied AI

- Scikit-learn
- Pandas
- NumPy
- Predictive Analytics
- Model Deployment
- Feature Engineering
- EDA
- Classification & Regression

ML Algorithms

- Linear Regression
- Logistic Regression
- Random Forest
- Gradient Boosting
- AdaBoost
- Decision Tree
- XGBoost
- ARIMA/SARIMA

Generative AI & NLP

- RAG (Retrieval-Augmented Generation)
- FAISS
- Vector Databases
- Semantic Search
- Context-Aware Retrieval
- Conversational AI
- Embedding Pipelines
- Prompt Engineering

Databases & Infrastructure

- MySQL
- MSSQL
- MariaDB
- Docker
- Git
- AWS S3
- Linux (Basic)
- CI/CD Basics

- Integrated Razorpay, PhonePe, CC Avenue, and logistics APIs including Shiprocket and DTDC.
- Worked on AWS S3-based media storage systems and scalable backend architecture.
- Improved backend performance and database efficiency through query optimization and modular development.

Saatirmind Software Services

May 2021 – July 2022

Laravel Developer

- Developed modular web applications using Laravel and MySQL.
- Worked on backend APIs, database optimization, and scalable application workflows.
- Improved system stability and deployment reliability through backend enhancements and debugging.

PROJECTS

Smart AI Assistant

Tech Stack: Laravel, Python, FAISS, Vector Database, RAG, NLP, LLM APIs

- Developed an AI-powered intelligent assistant with a custom Retrieval-Augmented Generation (RAG) pipeline for contextual question answering.
- Built complete AI workflows including:
 - Data collection and preprocessing
 - Embedding generation
 - FAISS vector indexing
 - Semantic retrieval pipeline
 - Context-aware response generation
- Implemented vector similarity search using FAISS for efficient document retrieval.
- Integrated bilingual conversational support, contextual AI interactions, and file attachment handling.
- Worked on prompt orchestration, retrieval optimization, and scalable backend AI integration.

Career Guidance Engine API

Tech Stack: FastAPI, Python, RAG, FAISS, Vector Database, NLP, JWT, Voice APIs

- Built an AI-powered career guidance platform delivering personalized recommendations and conversational career assistance.
- Developed a custom RAG pipeline with semantic retrieval and context-aware search workflows.
- Integrated FAISS indexing and vector similarity search for intelligent career information retrieval.
- Added voice assistant APIs for voice-enabled conversational interactions.
- Designed scalable FastAPI backend services with JWT authentication and modular architecture.

Security & Systems

- MFA
- eKYC Integrations
- Geofencing
- Secure Sessions
- Fraud Prevention Workflows
- SQL Injection Prevention

EDUCATION & CERTIFICATIONS

Executive Post Graduate Certification in Data Science & Artificial Intelligence

Intellipaat | 2025

Microsoft SQL Certification

Intellipaat | 2025

B.Tech – Information Technology

Dr. A.P.J. Abdul Kalam Technical University, Lucknow | 2020

Virtual Lab Development Program (Bootathon)

IIT Bombay | 2018

Industrial Equipment Monitoring & Predictive Maintenance System

Tech Stack: FastAPI, Streamlit, Python, Scikit-learn, Docker

- Engineered an AI-powered predictive maintenance platform for industrial equipment monitoring and fault prediction.
- Built preprocessing and feature engineering pipelines for sensor-based industrial datasets.
- Implemented Random Forest, XGBoost, Logistic Regression, and ensemble learning models for:
 - Fault Classification
 - Predictive Maintenance Forecasting
 - Equipment Failure Detection
- Developed FastAPI APIs and interactive Streamlit dashboards for monitoring, analytics, and explainability.
- Designed Docker-ready scalable architecture for production deployment.

Customer Churn Prediction Platform

Tech Stack: Python, Scikit-learn, Pandas, NumPy, XGBoost

- Developed an end-to-end customer churn prediction system using supervised machine learning techniques.
- Trained and evaluated multiple models including Logistic Regression, Random Forest, Gradient Boosting, AdaBoost, Decision Tree, and XGBoost.
- Achieved highest accuracy of 80.3% using Logistic Regression after comparative model evaluation.
- Built reusable ML pipelines for preprocessing, evaluation, and prediction workflows.

KEY STRENGTHS

- Backend + AI Engineering
- Product Ownership & System Design
- Fintech & Secure Transaction Systems
- FastAPI & Laravel Development
- AI Integration & RAG Pipelines
- Predictive Analytics & ML Systems
- Scalable API Architecture
- Problem Solving & Performance Optimization